

Eldridge Alexander

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SUMMARY

Security and IT professional with experience in building teams and programs to drive organizational security, profitability, and growth. Champions efficiency and focuses on developing and deploying solutions for a wide range of security and business challenges. Diverse experience in both scaling in smaller companies, as well as effectively leading teams in well established organizations.

SKILLS

- Technology & business strategy
- Bash
- Python, Django, Flask
- Git, CI/CD
- Enterprise IT
- BeyondCorp/Zero Trust
- Cloudflare, GCP, AWS
- SAML, OIDC, Okta, Keycloak
- IAM, WebAuthn
- Infrastructure
- Kvm, libvirt
- PKI, mTLS
- DNS, DoT, DoH
- Docker, Podman Kubernetes
- Wireguard, Tailscale

PROFESSIONAL EXPERIENCE

2023 - Present

Security Engineer, Hudson River Trading, New York, NY

Designed and implemented endpoint security systems and secure remote access

- Modernized infrastructure with a Zero Trust approach for SSO, adding device health info to authorization pipelines and removing default trust in many systems without negatively affecting user experience
- Transitioned majority of users to “passwordless” phish-proof authenticators (Webauthn/passkeys, Okta FastPass)
- Significantly reduced authentication prompts experienced by users while increasing security posture
- Consolidated various device health check processes into a client certificate issuance process, greatly simplifying administration, troubleshooting, and user experience
- Built automated application blocking process for endpoints in Python
- Architected encrypted and filtered endpoint DNS for globally distributed endpoint fleet
- Evaluated and built proof-of-concept implementations with various vendors
- Managed and administered multiple SaaS applications and vendor relationships used in securing the fleet
- Regularly advised our macOS, Windows, and Linux endpoint teams on security and fleet management
- Various endpoint and infrastructure automation using primarily Python and Bash

2022 - 2023

Security Architect & Consultant, Eldridge Alexander Consulting, New York, NY

Implemented industry security best practices and architected custom solutions .

- Analyzed client infrastructure, provided wide ranging security report with findings and remediation recommendations
- Recommended and architect Zero Trust approaches to application access
- Modernized network and VPN architecture
- Recommended, configured, and deployed security systems e.g. MDM, SSO
- Recommended, configured, and deployed security systems e.g. log aggregation and dashboard tools
- Created custom file sharing solution using Samba to meet compliance requirements

2021 - Present

Associate Director of Cloud Security, Gemini, New York, NY

Established the Cloud Security team.

- Collaborated in architecting Zero Trust based architecture for our internal services based on Okta’s offerings. (HTTPS traffic reverse proxied at our infrastructure edge with multiple data points for making Authn decisions)
- Drove account credential standardization (e.g. enforcing hardware MFA)
- Ensured seamless transition from legacy VPN access to Zero Trust access by careful DNS changes to legacy system
- Worked to ensure a seamless transition from legacy datacenter deployments to cloud, with my team assisted broader engineering in employing cloud native architecture where possible (E.g. AWS Network Firewall as opposed to host-based iptables)

2021

Information Security Manager, Kong Inc., Ann Arbor, MI

Initiated the Information Security Program

- Implemented process and software automation for common tasks, e.g. a Q&A database for our sales teams to use with potential sales leads’ security questionnaires.
- Established focus areas and prioritized work based on limited time, budget, and personnel

2018 - 2020

Manager of Duo Labs, Duo Security (a Cisco Company), Ann Arbor, MI

Defined and implemented processes for security research, and aligned it with business objectives. Facilitated integration of the team into Cisco after acquisition. Led architecting of next generation security systems, authentication, and authorization technologies.

- Created processes to effectively transfer research results into product features
- Led research into applying modern security to workloads focusing on Docker, Kubernetes, & Istio
- “Moved security left” by working with my team integrating security tooling into CI/CD pipelines
- Worked with team to architect a Zero Trust proof of concept for making arbitrary TCP traffic be able to be proxied through Duo’s Network Gateway using WireGuard or WebSockets.
- Led initiatives to build passwordless/WebAuthn tools and communicate them to the company
- Led user testing to ensure a “passwordless” WebAuthn workflow was accessible to non-engineers and that privacy concerns were addressed by UI.
- Guided team through Cisco’s acquisition created partnerships with similar teams at Cisco
- Wrote papers and presented talks both internally and publicly at industry conferences
- Presented at Black Hat and THOTCON conferences on encrypted DNS and 5G-IoT respectively

2016 - 2018

Head of IT Infrastructure, Cloudflare, San Francisco, CA

Built IT team from the ground up to support the fast growth of employees and tools. Drove the development of automated tooling and effective documentation to facilitate scale. Absorbed tools and shadow IT from other teams into IT team. Drove adoption of modern BeyondCorp/Zero Trust security methodology for IT tool deployment.

- Moved user-facing tools to employ Zero Trust security principles using Cloudflare infrastructure (now Cloudflare Access)
- Worked internally to use leanings about deploying BeyondCorp tools to develop what would become Cloudflare Access
- Built account creation/deletion automation using Python and Flask consolidating multiple systems, reducing onboarding time by 75% and making account deletion more reliable improving company security posture
- Absorbed various systems into IT, changed them to meet security and availability standards. This included our Atlassian suite, VPN concentrator, and Twilio+Cisco phone routing system.
- Drove documentation, metrics, and asset tracking improvement to meet IPO-readiness needs
- Automated regular tasks using Python and Bash, primarily our account systems and AV systems making them more accessible and self-service.
- Oversaw new office building deployments in multiple cities (San Francisco, Austin). Ensure quality network access, and other networking and physical security needs.
- Created internal short link system on GCP using Python
- Procured and deployed endpoint management systems to maintain and enforce security posture across all endpoints
- Created ticket “taxonomies” so as to have reliable metrics to use to determine resource allocation
- Worked cross-functionally to architect scalable directory services, as well as asset and user tracking to satisfy technical and auditing requirements

2014 - 2016

Operations Engineer, Google, Mountain View, CA

Responsible for a “Techstop” location and providing IT services to my region. Improved and automated common tasks for users in my region. Created and led training for peer and junior engineers on various technologies.

- Led Techstop team of 7 in supporting 7,000+ users
- Created App Engine app to securely and remotely reimage Linux workstations outside the corporate network using Python, iPXE, and Bash.
- Developed processes and signage to facilitate self service for device return, accessory requests, and 2FA enrollment
- Was responsible for creating and delivering substantial amounts of our departments training material for new employees as well as those looking for career growth including Security, Linux, Android, and ChromeOS.

2012 - 2014

Fieldtech, Google, Ann Arbor, MI

Responsible for providing IT services to Google’s global employees. Trained junior employees, and contributed to internal IT automation tooling.

- Worked on global team to remotely support all 100,000+ employees
- Trained new employees on internal tools
- Mentored teammates, particularly in Linux administration
- Assisted in creation of automated ticket routing rules, substantially reducing wait time for our end users
- Deployed code changes to our internal support tools to ease certain support interactions
- Created and presented TEDx talk on FOSS in education

2006 - 2012

Owner & Engineer, EZap Technology, Murfreesboro, TN

Business owner providing IT and AV consultant services to local SMB companies.

- Business IT support (e.g. email, networking, file servers, laptop endpoints)
- Designed, deployed, and maintained several websites for clients, including ecommerce offerings
- Performed video editing and production for clients
- Published writings in peer reviewed journals
- Worked with contractor to create a web app for mass texting on App Engine

2004 - 2012

Media Specialist, North Boulevard, Murfreesboro, TN

Responsible for providing IT, AV, and networking services to employees.

2010 - 2012

Sports TV Director, Middle Tennessee State University, Murfreesboro, TN

Responsible for directing live TV events that were broadcast on CSS or ESPN3.

2008 - 2009

Teacher, Cedar Hall School, Christiana, TN

Responsible for creating and delivering classes on basic technology concepts to middle and high schoolers.

PATENTS, PUBLICATIONS, & CONFERENCE PRESENTATIONS

[Identity services for passwordless authentication](#), US Patent Office

[Needing the DoH: The Ongoing Encryption and Centralization of DNS](#), Black Hat USA

[Remote Teams in Transition](#), Duo Blog

[Malicious Devices on Hostile Networks at Home](#), THOTCON

[How Technology is Changing Learning](#), TEDx Oakland

[Local Content Caching](#), TD Commons

[Credential Sharing](#), US Patent Office

[Operating System Interface for Credential Management](#), US Patent Office

[Google Docs in the Classroom](#), Journal for Computing Teachers

[Which Web Tool to Select](#), SITE Conference

EDUCATION

Middle Tennessee State University

Bachelor of Science, Major: Mass Communication, Minor: Theater -- *Buchanan Fellow, Honors Graduate*